

UNIVERSITÀ DEGLI STUDI DI TORINO

Corso di Laurea Magistrale in Matematica



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The ideally exact categories

A non-pointed generalization
of the notion of semiabelian category

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Introduction

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Chapter 1

Theory

1.1 Category over an object

Given an object E in a category $\mathcal{A}$, we call $(\mathcal{A} \downarrow E)$ the category of objects over E , we denote an object by $(A, \alpha : A \rightarrow E)$.

A morphism between two objects (A, α) and (A', α') is a morphism $f : A \rightarrow A'$ such that $\alpha' f = \alpha$.

1.2 Connected limits

Definition 1.2.1. A connected limit is a limit over a connected category. A category is connected if it is inhabited and any two objects can be connected by a zigzag of morphisms

$$\begin{array}{ccccccc}
 x_0 = y_1 & & & y_3 & & & y_n = x_1 \\
 & \swarrow & & \nwarrow & \swarrow & & \nwarrow \\
 & & y_2 & & & \dots & &
 \end{array}$$

where each symbol $\leftrightarrow$ is a morphism $y_i \rightarrow y_{i+1}$ or $y_{i+1} \rightarrow y_i$.

Example 1.2.2. Some examples

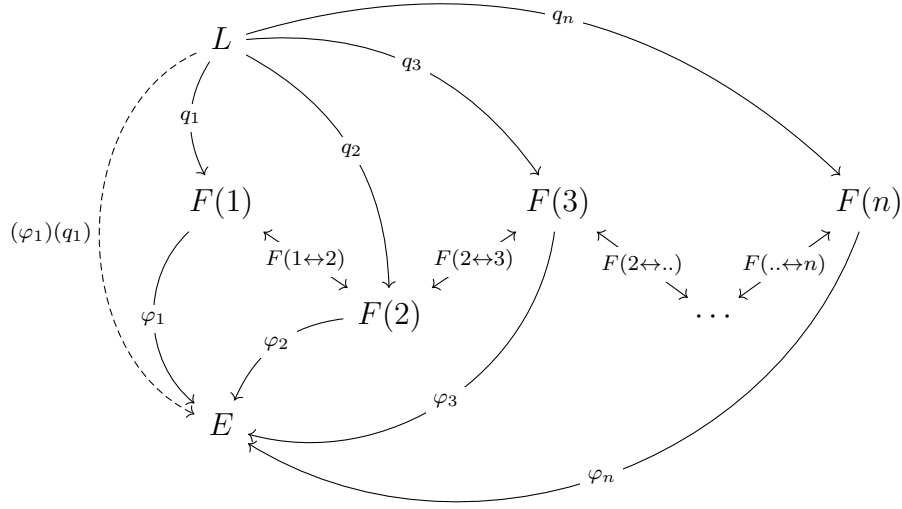
- The pullback is a connected limit.

$$\begin{array}{ccc}
 & 2 & \\
 & \downarrow j & \\
 1 & \xrightarrow{i} & 3
 \end{array}$$

- The equalizer is a connected limit.

$$1 \begin{matrix} \xrightarrow{i} \\ \xrightarrow{j} \end{matrix} 2$$

Remark 1.2.3. The connected limits in the category $(\mathcal{A} \downarrow E)$ can be computed in $\mathcal{A}$. In fact we can provide to the object L of the limit cone $(L, (q_i)_{i \in I})$ in $\mathcal{A}$ a morphism to form an object in $(\mathcal{A} \downarrow E)$. In the zigzag below we chose for example $(\varphi_1)(q_1)$, but we could use whatever morphism of the type $(\varphi_i)(q_i)$. The connection guarantees that all the morphisms $q_1 \dots q_n$ involved in the limit in $\mathcal{A}$ are also morphisms in $(\mathcal{A} \downarrow E)$, ie $(\varphi_i)(q_i) = (\varphi_1)(q_1) \forall i$.



By the previous result, the following holds.

Proposition 1.2.4. Let E be an object in $\mathcal{A}$. Then the forgetful functor $U : (\mathcal{A} \downarrow E) \rightarrow \mathcal{A}$ creates all connected limits that $\mathcal{A}$ has.

Note that this result is not true for all the limits. This is extensively shown in the following example.

Example 1.2.5. The product is not a connected limit, in fact as we can observe the following category is not connected.

$$1 \qquad 2$$

We try to compute the product directly in $(\mathcal{A} \downarrow E)$. Let (D, δ) and (D', δ') be

two objects in $(\mathcal{A} \downarrow E)$, the product is given by the following pullback

$$\begin{array}{ccccc}
 & & A \times_B A' & \longrightarrow & A' \\
 & \swarrow & & \searrow \alpha' & \\
 A & \xrightarrow{\alpha} & E & & \\
 & \searrow \alpha & \Downarrow & \swarrow \alpha' & \\
 & & E & &
 \end{array}$$

which doesn't correspond to the product $D \times D'$ in $\mathcal{A}$.

$$\begin{array}{ccccc}
& & A \times_B A' & \longrightarrow & A \\
& \swarrow & & \searrow \alpha' & \\
A & \xrightarrow{\alpha} & E & & \\
& \searrow \alpha & \Downarrow & \swarrow \alpha' & \\
& & E & &
\end{array}$$

which doesn't correspond to the product $D \times D'$ in $\mathcal{A}$.

1.3 The pullback functor and its adjunction

Given a category $\mathcal{A}$ with pullbacks and a morphism $p : E \rightarrow B$ in $\mathcal{A}$, one can consider functors pullback and shrink

$$\begin{array}{ccc} & \xrightarrow{p^*} & \\ (\mathcal{A} \downarrow B) & & (\mathcal{A} \downarrow E) \\ & \xleftarrow{p_!} & \end{array}$$

defined as follows:

- $p^* : (\mathcal{A} \downarrow E) \rightarrow (\mathcal{A} \downarrow B)$ is the pullback functor

The first diagram shows a pullback square:

$$\begin{array}{ccc} A & & E \times_B A \\ \downarrow \alpha & \xrightarrow{p^*} & \downarrow \text{proj}_1 \\ B & & E \end{array}$$

The second diagram shows a pushout square:

$$\begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \swarrow & & \searrow & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}'_2} & E \times_B A' \\ & \searrow \alpha & & \swarrow \alpha' & & & \\ & & B & & & & \\ & \swarrow \text{proj}_1 & \uparrow p & \searrow \text{proj}'_1 & & & \\ & & E & & & & \end{array}$$

- $p_! : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is the shrink functor

$$\begin{array}{ccc} D & & D \\ \downarrow \delta & \xrightarrow{p!} & \delta \downarrow \searrow p\delta \\ E & & E \xrightarrow{p} E \end{array}$$

$$\begin{array}{ccc}
 D & \xrightarrow{d} & D' \\
 \delta \downarrow & \nearrow \delta' & \\
 E & &
 \end{array}
 \xrightarrow{p_!}
 \begin{array}{ccc}
 D & \xrightarrow{d} & D' \\
 & \searrow p\delta & \downarrow p\delta' \\
 E & \xrightarrow{p} & B
 \end{array}$$

Proposition 1.3.1. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks, then $p_!$ preserves connected limits.*

Proof. By Remark 1.2.3 connected limits are computed in $\mathcal{A}$ and $p_!$ doesn't affect objects and morphism in $\mathcal{A}$. $\square$

Proposition 1.3.2. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks, then*

$$\begin{array}{ccc}
 & p^* & \\
 (\mathcal{A} \downarrow B) & \xrightarrow{\quad} & (\mathcal{A} \downarrow E) \\
 & p_! &
 \end{array}
 \quad \text{with } p_! \dashv p^*$$

is an adjunction. In particular the unit is given by

$$\begin{aligned}
 \eta : \text{Id}_{(\mathcal{A} \downarrow E)} &\rightarrow p^* p_! \\
 \eta_{(D, \delta)} &= \langle \delta, 1_D \rangle : (D, \delta) \rightarrow (E \times_B D, \text{proj}_1)
 \end{aligned}$$

$$\begin{array}{ccccc}
 D & & & & 1_D \\
 & \searrow \exists! \langle \delta, 1_D \rangle & & & \searrow \\
 & E \times_B D & \xrightarrow{\quad} & D & \\
 \delta \searrow & \downarrow & & \downarrow p\delta & \\
 & E & \xrightarrow{p} & B &
 \end{array}$$

and the counit is given by

$$\begin{aligned}
 \epsilon : p_! p^* &\rightarrow \text{Id}_{(\mathcal{A} \downarrow B)} \\
 \epsilon_{(A, \alpha)} &= \text{proj}_2 : (E \times_B A, p(\text{proj}_1)) \rightarrow (A, \alpha)
 \end{aligned}$$

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\text{proj}_2} & A \\
 \downarrow \text{proj}_1 & & \downarrow \alpha \\
 E & \xrightarrow{p} & B.
 \end{array}$$

Proof. Let us show that this is actually an adjunction. First we want to prove that the unit is a natural transformation. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc}
 D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D \\
 \downarrow g & & \downarrow p^* p_!(g) = \tilde{g} \\
 D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D'
 \end{array}$$

In order to do that we need to expand the action of the functor $p^* p_!$ on the morphism g . We have:

[illegible]

Considering the diagram

$$\begin{array}{ccccc}
D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D & & \\
\downarrow g & \searrow & \downarrow p^* p_!(g) = \tilde{g} & & \\
D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\
& & \downarrow \text{proj}'_1 & & \downarrow p\delta' \\
& & E & \xrightarrow{p} & B
\end{array}$$

we can prove the naturality in the left diagram by the universal property of the pullback in the right. Since $(\text{proj}_1, \text{proj}_2)$ are jointly monic, by the following compositions

- $(\text{proj}'_2)(\tilde{g})(\eta_{(D,\delta)}) = (g)(\text{proj}_2)(\eta_{(D,\delta)}) = (g)(1_D) = g$
- $(\text{proj}'_2)(\eta_{(D',\delta')})(g) = (1_{D'})(g) = g$
- $(\text{proj}'_1)(\tilde{g})(\eta_{(D,\delta)}) = (\text{proj}_1)(\eta_{(D,\delta)}) = \delta$
- $(\text{proj}'_1)(\eta_{(D',\delta')})(g) = (\delta')(f) = \delta$

we conclude that $(\tilde{g})(\eta_{(D,\delta)}) = (\eta_{(D',\delta')})(g)$. So the naturality condition of η is satisfied.

Now we want to prove that the counit is a natural transformation. Consider $f : (A, \alpha) \rightarrow (A', \alpha')$ a morphism in $(\mathcal{A} \downarrow B)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc} E \times_B A & \xrightarrow{\epsilon_{(A, \alpha)}} & A \\ p_! p^*(f) = \tilde{f} \downarrow & & \downarrow f \\ E \times_B A' & \xrightarrow{\epsilon_{(A', \alpha')}} & A' \end{array}$$

As similarly done above, we need to expand the action of the functor $p_!p^*$ on the morphism f , we have:

$$\begin{array}{ccc}
 \begin{array}{c} A \xrightarrow{f} A' \\ \downarrow \alpha \quad \swarrow \alpha' \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & & \swarrow \alpha' & & \searrow \text{proj}_1' & \\ & & B & & & & \\ & \swarrow \text{proj}_1 & & \nwarrow p & & \swarrow \text{proj}_1' & \\ & & E & & & & \end{array} \\ \exists! \tilde{f} \end{array} \\
 \\
 \xrightarrow{p!} & & \begin{array}{c} \begin{array}{ccccc} E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & & \swarrow \alpha' & & \searrow p(\text{proj}_1') & \\ & & B & & & & \\ & \swarrow p(\text{proj}_1) & & \nwarrow & & \swarrow & \\ & & E & & & & \end{array} \\ \exists! \tilde{f} \end{array}
 \end{array}$$

Since $\epsilon_{(A,\alpha)}$ is the $\text{proj}_{2,E \times_B A}$, the composition $(f)(\epsilon_{(A,\alpha)}) = (\epsilon_{(A',\alpha')})(\tilde{f})$ holds by the definition of $\tilde{f}$. So also the naturality of ϵ is satisfied.

Now we can check that the unit and counit fulfil the first triangle identities. Let (A, α) be an object in $(\mathcal{A} \downarrow B)$, we have:

$$((p^* \epsilon) \circ (\eta p^*))_{(A,\alpha)} = (p^* \epsilon)_{(A,\alpha)} (\eta p^*)_{(A,\alpha)} = p^* (\text{proj}_{2,E \times_B A}) (\eta_{(E \times_B A, \text{proj}_1)}) = \circledast$$

Considering first how p^* acts on the morphism $\text{proj}_{1,E \times_B A}$

$$\begin{array}{ccc}
 \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_{1,E \times_B A}} & A \\ & \searrow \alpha(\text{proj}_{2,E \times_B A}) & \swarrow \alpha \\ & & B \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} E \times_B (E \times_B A) & \xrightarrow{\text{proj}_{2,E \times_B (E \times_B A)}} & E \times_B A & \xrightarrow{\text{proj}_{2,E \times_B A}} & A & \xleftarrow{\text{proj}_{2,E \times_B A}} & E \times_B A \\ & \searrow (\alpha)(\text{proj}_{2,E \times_B A}) & & \swarrow \alpha & & \searrow \text{proj}_{1,E \times_B A} & \\ & & B & & & & \\ & \swarrow \text{proj}_{1,E \times_B (E \times_B A)} & & \nwarrow p & & \swarrow & \\ & & E & & & & \end{array} \\ \circledast = (\tilde{\text{proj}}_2)(\eta_{(E \times_B A, \text{proj}_1)}) \end{array}
 \end{array}$$

Finally we can check the second triangle identities. Let (D, δ) be an object in $(\mathcal{A} \downarrow E)$, we have:

$$\begin{aligned} ((\epsilon p!) \circ (p! \eta))_{(D, \delta)} &= ((\epsilon p!)_{(D, \delta)} (p! \eta))_{(D, \delta)} = (\epsilon_{p!(D, \delta)}) p! (\langle \delta, 1_D \rangle) = \\ &= (\epsilon_{(D, p\delta)}) p! (\langle \delta, 1_D \rangle) = \text{proj}_{2, E \times_B D} (\langle \delta, 1_D \rangle) = 1_D = \text{Id}_{(D, \delta)} \end{aligned}$$

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_{2, E \times_B D}} & D \\ \downarrow \text{proj}_{1, E \times_B D} & & \downarrow p\delta \\ E & \xrightarrow{p} & B \end{array}$$

So we can conclude that the adjunction holds. $\square$

Remark 1.3.3. p^* is a right adjoint functor, so it preserves limits. $p_!$ is a left adjoint functor, so it preserves colimits.

Property 1.3.4. The diagram constructed by p^* acting on a morphism $f : (A, \alpha) \rightarrow (A', \alpha')$ is a pullback in $(\mathcal{A} \downarrow B)$.

$$\begin{array}{ccc} E \times_B A & \xrightarrow{p^*(f)=\tilde{f}} & E \times_B A' \\ \downarrow \text{proj}_2 & & \downarrow \text{proj}'_2 \\ A & \xrightarrow{f} & A' \end{array}$$

Proof. We prove it in $\mathcal{A}$, then by Remark 1.2.3 it holds in $\mathcal{A} \downarrow B$. Since the right and the total diagrams below are pullback, then the left diagram is a pullback.

$$\begin{array}{ccccc} & & \text{proj}_1 & & \\ & \nearrow & & \searrow & \\ E \times_B A & \xrightarrow{p^*(f)=\tilde{f}} & E \times_B A' & \xrightarrow{\text{proj}'_1} & E \\ \downarrow \text{proj}_2 & & \downarrow \text{proj}'_2 & & \downarrow p \\ A & \xrightarrow{f} & A' & \xrightarrow{\alpha'} & B \\ & \searrow & & \nearrow & \\ & & \alpha & & \end{array}$$

$\square$

We need the following definition in order to claim some properties.

Definition 1.3.5. Let γ be a natural transformation between two functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, we say that γ is a cartesian natural transformation if for every morphism

$f : A \rightarrow B$ in $\mathcal{C}$, the following diagram is a pullback:

$$\begin{array}{ccc} F(A) & \xrightarrow{\gamma_A} & G(A) \\ \downarrow F(f) & & \downarrow G(f) \\ F(B) & \xrightarrow{\gamma_B} & G(B) \end{array}$$

ie the naturality diagrams are pullback.

Property 1.3.6. The unit η and the counit ϵ are cartesian.

Proof. First we prove that η is cartesian. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$. Applying *Property 1.3.4* to $p_!(g)$, we obtain that the following diagram is a pullback.

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_2} & D \\ \downarrow p^* p_!(g) = \tilde{g} & & \downarrow g \\ E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \end{array}$$

Considering the following commutative diagram, since the right diagram is a pullback, as we just proved, and the total diagram is a pullback, we can conclude that the left diagram is a pullback, as stated.

$$\begin{array}{ccccc} & & 1_D & & \\ & \nearrow & & \searrow & \\ D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D & \xrightarrow{\text{proj}_2} & D \\ \downarrow g & & \downarrow p^* p_!(g) = \tilde{g} & & \downarrow g \\ D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\ & \nwarrow & & \nearrow & \\ & & 1_{D'} & & \end{array}$$

Now we want to prove that ϵ is cartesian. Let $f : (A, \alpha) \rightarrow (A', \alpha')$ be a morphism in $(\mathcal{A} \downarrow B)$. By *Proposition 1.3.1* applied at the pullback in *Property 1.3.4* for f we obtain that the following diagram is a pullback.

$$\begin{array}{ccc} E \times_B A & \xrightarrow{\epsilon_{(A, \alpha)} = \text{proj}_2} & A \\ \downarrow p_! p^*(f) = \tilde{f} & & \downarrow f \\ E \times_B A' & \xrightarrow{\epsilon_{(A', \alpha')} = \text{proj}'_2} & A' \end{array}$$

□

Proposition 1.3.7. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks. Then p^* preserves coequalizer of kernel pairs whose coequalizers are pullback stable regular epimorphisms.*

Proof. Consider the following diagram

$$\begin{array}{ccccc}
 C & & & & C' \\
 \exists!s \downarrow & \nearrow r_2 & & \nearrow u & \exists!t \downarrow \\
 p^*(\text{Eq}(q)) & \xrightarrow[\tilde{p}_1]{\tilde{p}_2} & p^*(A) = A \times_B E & \xrightarrow{\tilde{q}} & p^*(\text{Coeq}(p_1, p_2)) \\
 \downarrow & & \downarrow & & \downarrow \\
 \text{Eq}(q) & \xrightarrow[p_1]{p_2} & A & \xrightarrow{q} & \text{Coeq}(p_1, p_2)
 \end{array}$$

Let q be the coequalizer of the kernel pair $\text{Eq}(q)$. We can affirm by *Property 1.3.4* applied to q that the diagram in the bottom-right corner is a pullback. Since by hypothesis q is a pullback stable regular epimorphism, we have that $\tilde{q}$ is a regular epimorphism, ie it is the coequalizer of some parallel arrows. Let (r_1, r_2) be such pair.

Now we want to prove that $\tilde{q}$ is also the coequalizer of $(\tilde{p}_1, \tilde{p}_2)$. Since $\tilde{q}$, $\tilde{p}_1$ and $\tilde{p}_2$ are obtained applying p^* to the kernel pair $\text{Eq}(q)$, and p^* preserves limits, then $p^*(\text{Eq}(q))$ is the kernel pair of $\tilde{q}$. So by the universal property there exists a unique morphism $s : C \rightarrow p^*(\text{Eq}(q))$ such that $\tilde{p}_1 s = r_1$ and $\tilde{p}_2 s = r_2$. Now let u be a morphism such that $u\tilde{p}_1 = u\tilde{p}_2$. We have that $ur_1 = u\tilde{p}_1 s = u\tilde{p}_2 s = ur_2$, so by the universal property of the coequalizer $\tilde{q}$ there exists a unique morphism $t : p^*(\text{Coeq}(p_1, p_2)) \rightarrow C'$ such that $t\tilde{q} = u$. So $\tilde{q}$ is also the coequalizer of $(\tilde{p}_1, \tilde{p}_2)$. $\square$

1.4 The monad associated to the adjunction

Consider the monad associated to the adjunction we just studied, we want to study some properties that will be useful.

Property 1.4.1. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks. The monad $(p^*p_!, \eta, p^*\epsilon p_!)$ is cartesian.*

Proof. We need to prove that the category $(\mathcal{A} \downarrow E)$ has pullback, the monad preserves pullbacks, and both the multiplication $p^*\epsilon p_!$ and the unit η are cartesian.

The category $(\mathcal{A} \downarrow E)$ has pullbacks, by *Proposition 1.2.4*, since $\mathcal{A}$ has. The monad $p^*p_!$ preserves pullbacks, in fact p^* is a right adjoint so it preserves limits, therefore preserves pullbacks, and $p_!$ preserves pullbacks by *Proposition 1.3.1*. We already proved that the unit η is cartesian, so it remains to prove that the multiplication $p^*\epsilon p_!$ is cartesian. Since ϵ is cartesian and p^* preserves pullbacks as shown above, we can conclude that the multiplication is cartesian. $\square$

Definition 1.4.2. A monad (T, η, μ) on a category $\mathcal{X}$ with finite coproducts is said to be essentially nullary if the morphism $[T(!_X), \eta_X]$ is a strong epimorphism for each object X in $\mathcal{X}$, where $!_X$ is the unique morphism from the initial object of $\mathcal{X}$ to X .

Now we claim a lemma which will be useful to prove some theorems.

Lemma 1.4.3. A category $\mathcal{X}$ is protomodular id and only if for each pullback

$$\begin{array}{ccc} B' \times_B A & \xrightarrow{k} & A \\ \updownarrow & & \downarrow p \updownarrow s \\ B' & \xrightarrow{\beta} & B \end{array}$$

the pair (k, s) is strongly epimorphic.

The proof is out of the scope of this thesis, but we can find it in (citazione)

Theorem 1.4.4. If $\mathcal{X}$ is a pointed protomodular category with finite coproducts and (T, η, μ) is a monad on $\mathcal{X}$ with cartesian η , then the monad is essentially nullary.

Proof. For each object X in $\mathcal{X}$ we can consider the following diagram:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & T(X) \\ \downarrow !^X \updownarrow !_X & & T(!^X) \updownarrow T(!_X) \\ 0 & \xrightarrow{\eta_0} & T(0) \end{array}$$

Since $\mathcal{X}$ is protomodular then by the lemma above, the pair $(\eta_X, T(!_X))$ is a strongly epimorphic. It is a standard fact that given f and g morphism, if (f, g) is a strongly epimorphic pair, then $[f, g]$ is a strong epi. So we conclude that $[\eta_X, T(!_X)]$ is a strong epi, ie the monad is essentially nullary. (citazione) $\square$